

tract-coreml — Op coverage matrix

This document accompanies the **tract-coreml** PR. It lists: 1. The canary corpus of models that drove translator development. 2. The complete op-family → MIL-op mapping for each translator. 3. Op classes that don't yet have a translator (gaps). 4. Op classes Apple's side has but we deliberately don't emit.

Source of truth in the repo: `coreml/docs/pr-prep.md` §§2–3.

2. Models tested (canary corpus)

13 model classes probed across CNN / segmentation / transformer / object-detection / super-resolution / audio / decoder-LLM. Each canary lives at `tract/coreml/tests/<model>_e2e.rs` with `#[ignore]`'d `*_coverage_probe` and (where applicable) `*_perf_*` tests.

Source-of-truth perf matrix lives in `pr-prep-perf-matrix.md` — generated by a measurement run (CoremlOp counts, first-run latency, steady-state RTF, CPU baseline where available, ANE engagement). The summary below is the structural picture.

2.1 Image classifiers (CNN)

(First-run = cold compile cache, single inference. Steady-state in §5.)

Canary	ONNX nodes	CoremlOps	Stranded	First-run	Steady-state	ANE engages?	Notes
MobileNet v2	533	1	0	8.84 ms	2.22 ms	185 mW sustained	First ANE engagement; 191× speedup over tract CPU.
SqueezeNet 1.x	213	1	0	5.41 ms	0.46 ms	Too small for ANE scheduler	50× speedup over tract CPU.
InceptionV3	1007	2	1× Ein-Sum	18.98 ms	12.18 (default All)	2.76× CPU→ANE inference	Pure CNN with 9× SumPool; Same-Upper-padding fix unblocked. 34× speedup over tract CPU at default All (post Re-size-F16 patch).

2.2 Segmentation / matting

Canary	CoremlOps	Stranded	First-run	Steady-state	ANE en- gages?	Notes
MODNet @ 512 ²	1	0	22.06 ms	10.11 ms	200– 600 mW (370 mW mean)	RTF 0.59 @60fps (1.7× head- room). 241× speedup over tract CPU (post Resize-F16 patch).
MODNet @ 256 ²	1	0	(not re-measured)	~5.9 ms (prior baseline)		RTF 0.35 @ 60 fps; 2.8× head- room.
RVM (re- current video matting)	1	0	15.10 ms	4.53 ms (de- fault All)	ANE rejects re- current loop	All 4 modes work post- ceil_mode fix. 111× speedup over tract CPU (post Resize- F16 patch). Important caveat: the origi- nal “RVM doesn’t engage ANE” claim was confirmed by per- mode la- tency compari- son (CPU+ANE only 1.59× faster than CPU-only — too small to be ANE accelera- tion). The recurrent- state in- put pat- tern trig- gers ANE

2.3 Object detection

Canary	CoremlOps	Stranded	First-run	Notes
YOLOv8n	7	7× rank-5 BN-bias Add + 3× Cast + 2× DFL EinSum	44.13 ms	Conv rank-3 abort fix unblocked. NMS not in default Ultralytics export — use nms=True to exercise NMS coverage.

2.4 Super-resolution / generative

Canary	CoremlOps	Stranded	First-run	Notes
Real-ESRGAN x4	3	4 (upsample chain — 2× AddAxis + 2× MultiBroadcastTo)	135.82 ms	LeakyRelu translator absorbed 279 ops; 100% Conv coverage (351/351). Was 375 CoremlOps / 55 s transform — now 3 CoremlOps / 6.7 s transform. Steady-state 131.24 ms = 12.7× over tract CPU (post Resize-F16 patch); the modest ratio reflects the upsample-chain CPU residuals (I14).

2.5 Transformer (image)

Canary	CoremlOps	Stranded		First-run	Notes
SAM 2 Hiera-Tiny encoder	39	24×	rank-5 BinOp + 10× Cast/Mul/Add residuals + 1× rank-6 BinOp/ Cast	1.82 s	Windowed-at- tention high- rank residuals; ANE compile fails (MILCompilerForANE FAILED 3× dur- ing compile).
ViT-base	1 	0	(probe doesn't run in- ference)		Full collapse (Concat const- input fix + general_matmul defer removal + Gather trans- lator). Steady- state TBD.

2.6 Audio (speech recognition)

Canary	CoremlOps	Stranded	First-run	Notes
DFN3 decoder (NNEF)	6	6× EinSum inside GRU tract_core_scan bodies + 2× Scan + 1× Cast	71.85 ms	RTF 0.070 (14× faster than real-time at 1 s chunks). Tract CPU 75.34 ms vs CoreML 71.37 ms = 1.06× speedup — quantifies the cost of the GRU Scan loops staying on CPU; CoreML only accelerates the outside-scan compute (~5% of wall time). Was 3 CoremlOps in prior baseline — see §5.1 fragmentation regression. Recurrent compute stays on CPU; needs RNN-preservation upstream work or pattern-detector translator (see tract-rnn-preservation-handout.md).
Parakeet decoder	12	8× LSTM-scan EinSums + 1× Gather + 2× Scan + 1× Cast	(not measured)	LSTM-Scan-bound (same architectural issue as DFN3).
Parakeet joint	2	3× M-less-degenerate EinSums + 3× Cast	(not measured)	“M-less” einsum (rank-1 output, no M axis) needs separate handling — see I12 in upstream-feedback.
Parakeet pre-processor	3	7× ops (Cast/ Eq/GTE/LT/ Iff/Add/Sub on	(not measured)	TDim-arithmetic chain; structural CPU-
Neon pre-processor	10	TDim, shape arithmetic) kernel (see 8	(not measured)	only, BArchitecturally parallel blocks (see upstream-feedback).

2.7 Decoder LLM

Canary	CoremlOps	Stranded	First-run	Notes
SmolLM2-135M	1 	2× Cast(I64) (RoPE table indexing)	(probe doesn't run inference)	Full collapse achieved this session for 30 transformer blocks + 61× Concat KV-cache + 60× RmsNorm + 30× Softmax + 30× Silu + 272× EinSum projections. ANE compile fails (MILCompilerForANE FAILED 1× — same symptom as SAM 2). Runs on GPU/CPU.
Phi-3-mini-4k	aborts at load	n/a	n/a	tract-onnx has no com.microsoft operator domain support — see B8 in upstream-feedback.

2.8 Models attempted but blocked at load

Model	Block	Reference
MediaPipe Selfie	tract symbolic-batch unification (B2)	tests/mediapipe_selfie_e2e.rs ([ignore]‘d)
Microsoft Phi-3-mini-4k ONNX (cuda-fp16)	com.microsoft domain operators (B8)	tests/phi3_e2e.rs
Granite 4.1 family	No public ONNX; only PyTorch + MLX. Would require optimum-cli conversion which surfaces Mamba (B8-style situation)	(no test file; investigated only)

Summary stats

- **13 model classes / 16 distinct probes**, covering CNN classifier / segmentation / video matting / object detection / SR / transformer encoder / windowed-attention transformer / NNEF audio / decoder LLM
- **6 models collapse to 1 mega-CoremlOp**  (MobileNet, SqueezeNet, MODNet, RVM, ViT, SmolLM2)
- **4 models translate but fragment** (InceptionV3 / Real-ESRGAN / YOLOv8 / SAM 2)

- **3 models structurally LLM/ASR-blocked** (DFN3 / Parakeet / Nemotron decoder+joint — recurrent Scan bodies stay on CPU)
 - **3 ANE-engaging canaries** (MobileNet, MODNet, InceptionV3)
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3. Op coverage matrix (translators)

28 translators across `tract/coreml/src/ops/*.rs` (excluding `mod.rs` and `rank.rs-helper`). Each handles one tract op family or one MIL op family.

3.1 Translators by file

3.2 Cross-cutting infrastructure

File	Role
<code>mod.rs</code>	Module registry + shared <code>const_tensor</code> helper that walks <code>Cast(Const)</code> chains.
<code>rank.rs</code>	<code>pad_to_rank(shape, target) / pad_to_rank_4 / try_strip_to_rank5</code> — single source of truth for the in-MLPackage rank convention. Used by every translator.

3.3 Op classes with NO translator yet (gaps)

Surveyed across all canary probes. Each tract op below would unblock a specific model class:

tract op	Where it surfaces	Estimated effort
<code>tract_core_scan</code> body recursion	DFN3 decoder, Parakeet decoder, Nemotron decoder	Architecturally invasive (~600 LoC pattern detector OR upstream tract change — see <code>tract-rnn-preservation-handout.md</code>)
Detection-specific: <code>NonMaxSuppression</code> , <code>TopK</code> , <code>ScatterND-with-NMS-shape</code>	YOLOv8 with <code>nms=True</code> export, DETR-family	~200 LoC each; not yet exercised by any canary
<code>Range</code>	Parakeet/Nemotron preprocessors (3× / 2× per model)	~50 LoC; integer-only iota
<code>BitNot</code>	Parakeet/Nemotron preprocessors	trivial via <code>mb.logical_not</code>
<code>STFT</code>	Parakeet/Nemotron preprocessors	~300 LoC; would unlock pure-audio preprocessing
<code>tract_core_bitnot</code> , <code>tract_core_eq</code> for TDim shape arithmetic	Parakeet/Nemotron preprocessors	Defer to upstream B5/B6 — these are TDim-shape ops that shouldn't reach a backend

3.4 Op classes Apple-side has but we deliberately don't emit

MIL op	Why we don't	Why we might revisit
<code>mb.scaled_dot_product_attention</code>	tract decomposes attention into <code>MatMul + Softmax + MatMul</code> — <code>general_matmul</code> handles it without needing the fused op	When the upstream RNN-preservation work (I16 — fused-contrib-op preservation) lands, the SDPA-fused form would benefit ANE engagement on transformers
<code>mb.gru</code> , <code>mb.lstm</code> , <code>mb.rnn</code>	tract surfaces RNN as <code>Scan(body=cell)</code> — pattern would have to detect cell-body shape	Same — see RNN preservation handout